

Sundar Viswanathan

DATA SCIENCE · MACHINE LEARNING · ARTIFICIAL INTELLIGENCE

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EDUCATION

National University of Singapore

MASTER'S IN ARTIFICIAL INTELLIGENCE

Singapore

Aug. 2018 - Jan. 2020

Amrita Vishwa Vidyapeetham

BACHELOR'S IN COMPUTER SCIENCE

Coimbatore, India

Aug. 2011 - July 2015

SKILLS

Programming Proficient: Python; Familiar: JAVA, C++, C, J2EE, PHP, C#, JavaScript, AngularJS, NodeJS.

Libraries

Pandas, scikit-learn, Keras, TensorFlow, PyTorch, Hugging Face Transformers, LangGraph, LangChain, CrewAI, NumPy, SciPy, FastAPI, Matplotlib, Streamlit, Anomalib, AllenNLP, spaCy, NLTK, fairseq, OpenCV, PIL, PCL, statsmodels, lifelines, PyOD, librosa, pyAudioAnalysis, reliability, Skforecast, Seaborn, Vuforia.

Tools

PowerBI, Tableau, GitHub Copilot, Jira, Confluence, Grafana, Gurobi, Mendix, DbVisualizer, Protégé, Postman, Unity3D, MATLAB.

DevOps

GitLab, Jenkins, Docker, NGINX.

Databases

PostgreSQL, InfluxDB, Blazegraph, Neo4j, GraphDB, ChromaDB, Hive, MySQL, SQLite.

Cloud

AWS, Microsoft Azure, GCP.

Languages

English, Tamil.

WORK EXPERIENCE

Philips

DATA SCIENCE & ANALYTICS SPECIALIST

Singapore

Dec 2024 - Present

- Leading regional data and AI initiatives for the Hospital Patient Monitoring team to enhance clinical decision-making and drive commercial growth across markets beyond North America, Europe, and China.
- Developing a pilot app for Clinical Application Specialists to reduce hospital alarm fatigue using historical alarm data and LLM-based AI insights.
- Building a consumables analytics app for the sales team to identify potential new opportunities from sales and installed base data.
- Developed a budgeting tool to simplify lifecycle solution quoting and speed up customer engagement.
- Designed a business performance dashboard for the ASEAN KR team to track KPIs and regional metrics.
- Built a partner performance dashboard to monitor and improve channel partner effectiveness across multiple markets.

Siemens Digital Industries

SENIOR DATA SCIENTIST

Singapore

July 2022 - Dec 2024

- Part of ASEAN IoT services team, focusing on industrial data science use cases.
- Built a pilot RAG-based digital assistant for a Siemens application to resolve L1 tickets, reducing resolution time and improving user satisfaction.
- Developed forecasting models and dashboards for the FIN team, enabling more accurate financial planning and faster decision-making.
- Led a development team to build a QRQC-focused information management system for an automobile manufacturer.
- Developed computer vision modules for industrial visual inspection use cases, boosting accuracy and reducing manual work.
- Served as ASEAN service owner for Senseye, a scalable predictive maintenance solution, driving adoption and reducing unplanned downtime.
- Applied data-driven techniques to improve chiller plant efficiency, resulting in energy savings and operational cost reduction.

Siemens Mobility

DATA SCIENTIST

Singapore

Sep. 2021 - June 2022

- Associated with the Rail Enterprise Asset Management System (REAMS) data analytics team. Responsible for reliability modules in REAMS.
- Developed long-term failure rate forecast models with deterioration effects for renewal planning.
- Built statistical models for analyzing the effect of predictive maintenance and shifting preventive maintenance on reliability metrics.

Siemens Advanta Consulting

DATA SCIENTIST

Singapore

June 2019 - Aug. 2021

- Worked on pilot projects and proof of concepts in the field of advanced analytics and machine intelligence.
- Developed a dialysis machine fault detection system with acoustic and vibration sensors for a leading healthcare service provider. Exploited data-centric approaches and self-supervised representation learning to recognize failures.
- Implemented an agent to summarize contracts utilizing language models and few-shot learning.
- Developed computer vision modules involving object matching, optical flow, and video streaming to achieve smart aquaculture.

GetVu

Co-FOUNDER & CTO

Bangalore, India

July 2015 - Mar. 2018

- GetVu is a software solutions provider for smart glasses like Epson BT200 and Microsoft HoloLens.
- Developed the initial version of a smart glass augmented reality solution called PikVu for improving the warehouse picking process.
- Developed applications for Microsoft HoloLens, Google Tango, Leap Motion, Intel Recon Jet Pro, Sony SmartEyeGlass, ODG R6 & R7, Epson BT200, RealWear HMT-1, Oculus Rift and Google Cardboard.

ACHIEVEMENTS

2022	Botbuilder Hackathon Winner , Siemens Chatbot Hub	Singapore
2021	Werner von Siemens Award - Spotlight , Siemens	Singapore
2015	Selected for River Accelerator II , Rothenberg Ventures	San Francisco, USA
2015	Outstanding Student Award , Amrita Vishwa Vidyapeetham	Coimbatore, India
2015	TIDE Innovation Award , Amrita TBI	Bangalore, India
2014	Selected for LEAP AXLR8R , SOSV	San Francisco, USA

SELECTED PROJECTS

PAAM - Personal AI Asset Manager

INDEPENDENT WORK

Jan 2026 - Present

- A multi-agent system that tracks and consolidates your investments.
- Provides real-time market insights, and keeps you informed - enabling seamless, efficient portfolio management.

Twiloc - Location Prediction Using Tweets

NATIONAL UNIVERSITY OF SINGAPORE

April 2019 - June 2019

- TwiLoc investigates the feasibility of geographically locating Twitter users based solely on tweet content.
- TwiLoc exploits deep learning techniques like LSTM and RNN for understanding the dialect differences across geographies.

SELECTED PUBLICATIONS

2020	Language Model like Pre-Training for Acoustic Data , Towards Data Science
2019	Entity Linking: A primary NLP task for Information Extraction , Analytics Vidhya
2019	Deep Learning for Survival Analysis , Towards Data Science

SELECTED COURSES

NATIONAL UNIVERSITY OF SINGAPORE

Neural Networks and Deep Learning | AI Planning and Decision Making | Knowledge Discovery and Data Mining | Data Fitting and Optimization Algorithms | Text Mining | Uncertainty Modelling in AI | Visual Computing | Intelligent Systems and Techniques | Distributed Systems

ONLINE

MLOps Specialization, deeplearning.ai, Coursera | Generative AI with Large Language Models, AWS, Coursera | Deep Learning Specialization, deeplearning.ai, Coursera | Mining Massive Datasets, Stanford, edX | Data Visualization Specialization, UCDavis, Coursera | Machine Learning, Stanford, Coursera | Web of Data, EIT Digital, Coursera | Agile Project Management, Google, Coursera

ADDITIONAL INFORMATION

2022	Professional Membership MSCS , Singapore Computer Society	Singapore
2019	Contributing Writer , Towards Data Science & Analytics Vidhya	Singapore
2015	Presented PikVu , IEEE Digital Senses Initiative Conference	San Francisco, USA
2014	Exhibited GetVu's AR Headset , Augmented World Expo	San Francisco, USA
2014	HackerRank Campus Ambassador , Amrita Vishwa Vidyapeetham	Coimbatore, India